SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes
Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

A traffic monitoring system requires **accurate motion direction detection** and **robust performance in dense traffic conditions**. The two approaches have different strengths.

Factor	Optical Flow	Motion Estimation
Direction detection	**Highly accurate**	Moderate
Detail	**High**	Lower
Dense traffic	Can be noisy	**More stable**

Factor	Optical Flow	Motion Estimation
Reliability	Lower in crowded scenes	**Higher**
Best use	Sparse/clear traffic	**Dense traffic**

Evaluation

Optical Flow:
Optical Flow calculates the movement of individual pixels between consecutive frames. It provides detailed information about the **direction and speed of moving vehicles**. However, when traffic is very dense, vehicles may overlap or move close together, producing noisy or confusing motion vectors.

Motion Estimation:
Motion Estimation focuses on estimating overall motion between frames. It provides **less detailed information** than Optical Flow but is more stable when many vehicles are present. This makes it less affected by overlapping objects and complex motion patterns.

Scenario-Based Decision

Consider a **highway during peak hours**, where many vehicles are moving close together. Optical Flow may generate many motion vectors from different vehicles, causing noise and making direction detection unreliable.

In the same situation, Motion Estimation can provide a more stable representation of the overall vehicle movement. Although it may not capture every small movement, its stability makes it more reliable for continuous traffic monitoring.

Conclusion

For **high-density traffic conditions, Motion Estimation is the more suitable approach** because robustness and stable tracking are more important than highly detailed motion information.

Optical Flow → Better accuracy and detail in clear/sparse traffic.
Motion Estimation → Better stability and robustness in dense traffic.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

A **Gaussian Mixture Model (GMM)** is used to model the background of a video and separate moving objects from static regions.

Scenario	GMM Accuracy	Requirement	Result
Static scenes	**90%**	≥ 80%	✅ Meets

Scenario	GMM Accuracy	Requirement	Result
Dynamic scenes	**65%**	≥ 80%	❌ Fails

Evaluation
Static Scenes:
GMM performs well with **90% accuracy**, which is above the required 80%. In a static environment, the background remains relatively unchanged, so GMM can effectively identify moving objects.
Dynamic Scenes:
GMM accuracy drops to **65%**, which is significantly below the required 80%. In dynamic environments, background elements such as **moving trees, shadows, changing illumination, or camera movement** can be incorrectly detected as foreground objects. This reduces the accuracy.
Does GMM Alone Meet the Requirement?

No. ❌
Although GMM satisfies the requirement in static scenes, it fails in dynamic scenes. Since the system must maintain **at least 80% accuracy across all scenarios**, GMM alone cannot meet the overall requirement.
Decision
GMM should **not be used alone**. It can be combined with additional techniques such as **adaptive background modeling, shadow removal, noise filtering, or motion-based detection** to improve performance in dynamic scenes.
Conclusion
GMM alone → Not suitable ❌
GMM + additional dynamic-scene handling → Recommended ✅
The decision is based on the fact that the **minimum required accuracy must be achieved in every scenario**, and GMM achieves only **65% in dynamic scenes**.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

Stereo Vision: high accuracy, needs two cameras
Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.
The system has two important constraints:

- **Only one camera is available**
- **Moderate depth accuracy is acceptable**

Comparison

Factor	Stereo Vision	Structure from Motion (SfM)
Cameras required	Two cameras	**One camera**
Accuracy	High	Moderate
Hardware requirement	High	**Low**
Cost	Higher	Lower
Meets constraints?	❌ No	✅ Yes

Evaluation

Stereo Vision:
Stereo Vision uses **two cameras** placed at different positions to estimate depth from the difference between their views. It provides high accuracy, but it cannot be directly used because the system has **only one camera**. Therefore, it violates the hardware constraint.

Structure from Motion (SfM):
SfM uses a **single camera** and multiple images captured from different viewpoints. It estimates camera movement and reconstructs the depth of the scene. Its accuracy is moderate, which is acceptable according to the requirement.

Constraint-Based Decision

Since **only one camera is available**, Stereo Vision is not feasible regardless of its higher accuracy. SfM satisfies both constraints:

- **Single camera → Satisfies hardware constraint** ✅
- **Moderate accuracy → Satisfies accuracy requirement** ✅

Conclusion

Structure from Motion (SfM) is the best and most feasible choice. It meets the hardware limitation and provides sufficient depth accuracy. Stereo Vision would provide better accuracy, but it cannot be selected because it requires **two cameras**.

6. Retail Motion Tracking Logic

A retail system requires:

- Accurate tracking when crowd density is low
- Stable tracking when density is high

Observations:

Optical Flow: accurate in low density, noisy in high density

Motion Models: stable in high density, less detailed

Compare both and evaluate a suitable approach or combination. Justify your reasoning logically.

The retail system has two requirements: **high tracking accuracy in low-density crowds** and **stable tracking in high-density crowds**.

Factor	Optical Flow	Motion Models
Low-density tracking	**Highly accurate**	Less detailed

Factor	Optical Flow	Motion Models
High-density tracking	Noisy	**Highly stable**
Motion details	High	Moderate
Overall suitability	Good for low density	Good for high density

Evaluation

Optical Flow:
Optical Flow tracks pixel-level movement and provides detailed information about object motion. When there are fewer people in the store, it can accurately track individual customers. However, when the crowd becomes dense, overlapping people and complex movements create noisy motion information.

Motion Models:
Motion Models provide a more stable estimate of movement, especially when many people are present. They are less affected by overlapping objects and crowded scenes. However, they provide less detailed motion information compared with Optical Flow.

Suitable Approach

A **combination of both methods** is the best solution.

- In **low-density conditions**, use **Optical Flow** for accurate and detailed customer tracking.
- In **high-density conditions**, switch to **Motion Models** for stable tracking.
- The system can use **crowd density detection** to automatically select the appropriate method.

Conclusion

Using only one method would not satisfy both requirements effectively. Therefore, a **hybrid approach combining Optical Flow and Motion Models** is recommended. It provides **high accuracy in low-density areas and stable tracking in high-density areas**, making it more reliable for a retail environment.

7. Defect Detection System Decision

Requirements:

- Detect subtle defects (>95% accuracy)
- Adapt to product variations

Options:

Traditional methods: fast, less adaptive
Deep learning: accurate, requires training

Compare both approaches and evaluate which meets requirements. Justify using performance vs adaptability logic.

The defect detection system must satisfy two main requirements:

- **Accuracy greater than 95%** for detecting subtle defects.
- **Adaptability** to different product types and variations.

Comparison

Factor	**Traditional Methods**	**Deep Learning**
Accuracy	Moderate	**Very high**
Subtle defect detection	Less effective	**Highly effective**
Adaptability	Low	**High**
Training required	Little/none	Required
Processing speed	**Fast**	Generally slower
Product variations	Less adaptable	**More adaptable**

Evaluation

Traditional Methods:
Traditional image processing techniques such as **thresholding, edge detection, and morphological operations** are fast and simple. However, their performance depends heavily on fixed rules and parameters. When product appearance, lighting, size, or texture changes, these methods may fail to detect defects correctly. Therefore, they are less suitable for subtle defects and varying products.

Deep Learning:
Deep learning models can learn complex patterns directly from training images. They are better at identifying **small or subtle defects** and can adapt to different product appearances when the training dataset contains sufficient variations. Although training requires more computational resources and a large labeled dataset, the resulting model can provide much higher accuracy.

Decision

Deep Learning is the better choice because the system requires **more than 95% accuracy** and must adapt to product variations. Traditional methods are faster, but their limited adaptability makes them unsuitable for these strict requirements.

Conclusion

Traditional methods → Fast but less accurate and less adaptive.
Deep Learning → More accurate and adaptable but requires training.
Therefore, **Deep Learning should be selected** because **accuracy and adaptability are the primary requirements**, while the additional training cost is acceptable.

15. Face Recognition System Design

Constraints:

- Accuracy ≥ 90%
- Works in low light
- Moderate computation

Options:
Traditional: 82%
Deep Learning: 93%
Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

The face recognition system has three main requirements:

- **Accuracy ≥ 90%**
- **Good performance in low-light conditions**
- **Moderate computational requirement**

Comparison

Method	Accuracy	Low-Light Performance	Computation	Meets Requirement?
Traditional	82%	Poor	Low	❌
Deep Learning	**93%**	**Good**	High/Moderate	✅
Hybrid	89%	Moderate	Moderate	❌

Evaluation

Traditional Method:
Traditional face recognition methods require relatively low computation, but the accuracy is only **82%**, which is below the required 90%. They are also more sensitive to lighting changes. Therefore, they are not suitable.

Deep Learning:
Deep learning achieves **93% accuracy**, exceeding the required 90%. It can learn robust facial features and can be trained using low-light images, making it more suitable for difficult lighting conditions. Although it generally requires more computation, it can be optimized using techniques such as **model compression or hardware acceleration** to achieve moderate computational requirements.

Hybrid Method:
The hybrid method requires moderate computation and may perform better under varying conditions, but its accuracy is only **89%**, which fails the 90% requirement.

Conclusion

Deep Learning is the best choice because it is the **only option that meets the required accuracy of ≥90%** and can be designed to work effectively in low-light conditions.

Although its computational requirement is higher, optimization can reduce the computational burden.

Final Decision: Deep Learning → Best choice ✅
Reason: It satisfies the most critical requirements of **high accuracy and low-light robustness**, with computation that can be optimized for deployment.

19. Video Analytics Pipeline

Constraints:

- Accuracy ≥ 90%
- Processing time < 100 ms

Options:
Pipeline A: 88%, 80 ms
Pipeline B: 91%, 120 ms
Pipeline C: 90%, 95 ms

Compare and evaluate the best pipeline. Justify your decision.

The system has two strict constraints:

- **Accuracy ≥ 90%**
- **Processing time < 100 ms**

Pipeline	Accuracy	Processing Time	Meets Requirements?
A	88%	80 ms	❌ No – accuracy too low

Pipeline	Accuracy	Processing Time	Meets Requirements?
B	91%	120 ms	❌ No – processing time too high
C	90%	95 ms	✅ Yes – meets both

Evaluation

Pipeline A:
It is fast, with a processing time of **80 ms**, but its accuracy is only **88%**, which is below the required 90%. Therefore, it cannot be selected.

Pipeline B:
It provides **91% accuracy**, satisfying the accuracy requirement. However, its processing time is **120 ms**, which exceeds the 100 ms limit. Therefore, it also fails the constraints.

Pipeline C:
It provides exactly **90% accuracy** and takes **95 ms** to process. Both values satisfy the given requirements. Although it has less accuracy than Pipeline B, it is the only pipeline that satisfies **both constraints simultaneously**.

Conclusion

Pipeline C is the best choice because it meets both required conditions:

Accuracy = 90% ≥ 90% ✅
Processing time = 95 ms < 100 ms ✅

Therefore, **Pipeline C should be selected** for the video analytics system.

26. Video Surveillance Upgrade

Constraints:

- Improve accuracy
- Maintain same latency

Options:
Model A: 85%, 50 ms
Model B: 90%, 50 ms

Compare and evaluate. Justify your decision.

The main requirements are to **increase accuracy** while **maintaining the same latency**. Therefore, the model with higher accuracy and equal processing time should be selected.

Model	**Accuracy**	**Latency**	**Evaluation**
A	85%	50 ms	Lower accuracy
B	**90%**	**50 ms**	Higher accuracy

Evaluation

Model A:
Model A provides **85% accuracy** with a latency of **50 ms**. Although its processing speed is acceptable, it does not provide the required improvement in accuracy compared with Model B.

Model B:
Model B achieves **90% accuracy** while maintaining the same **50 ms latency**. This means the surveillance system can improve its detection performance without increasing processing delay.

Decision

Model B is the better choice. It improves accuracy from **85% to 90%**, a **5 percentage-point improvement**, while keeping latency unchanged at **50 ms**.

Conclusion

Model B → Recommended ✅

It satisfies both objectives: **higher accuracy + same latency**. Therefore, Model B is the most suitable upgrade for the video surveillance system.